Princess Sumaya University for Technology
Communication Engineering Department
Applied Electromagnetics
Worksheet 2-Solution

1.a. Expressing $\boldsymbol{B}$ in Cartesian and Cylindrical coordinates. Using the equation:

$$\begin{bmatrix} \boldsymbol{A_x} \\ \boldsymbol{A_y} \\ \boldsymbol{A_z} \end{bmatrix} = \begin{bmatrix} \sin\theta\cos\phi & \cos\theta\cos\phi & -\sin\phi \\ \sin\theta\sin\phi & \cos\theta\sin\phi & \cos\phi \\ \cos\theta & -\sin\theta & 0 \end{bmatrix} \begin{bmatrix} \boldsymbol{A_r} \\ \boldsymbol{A_\theta} \\ \boldsymbol{A_\phi} \end{bmatrix}$$

Then

$$\begin{bmatrix} \boldsymbol{B_x} \\ \boldsymbol{B_y} \\ \boldsymbol{B_z} \end{bmatrix} = \begin{bmatrix} \sin\theta\cos\phi & \cos\theta\cos\phi & -\sin\phi \\ \sin\theta\sin\phi & \cos\theta\sin\phi & \cos\phi \\ \cos\theta & -\sin\theta & 0 \end{bmatrix} \begin{bmatrix} \frac{\mathbf{10}}{\mathbf{r}} \\ \mathbf{r}\cos\theta \\ \mathbf{1} \end{bmatrix}$$

or

$$B_x = \frac{10}{r}\sin\theta\cos\phi + rcos^2\theta\cos\phi - \sin\phi$$

$$B_y = \frac{10}{r}\sin\theta\sin\phi + rcos^2\theta\sin\phi + \cos\phi$$

$$B_z = \frac{10}{r}\cos\theta - r\cos\theta\sin\theta$$

but $r = \sqrt{x^2 + y^2 + z^2}$

$$\theta = tan^{-1}\frac{\sqrt{x^2 + y^2}}{z}$$

and $\phi = tan^{-1}\,(\frac{y}{x})$

Hence

$$\sin\theta = \frac{\rho}{r} = \frac{\sqrt{x^2+y^2}}{\sqrt{x^2+y^2+z^2}}$$

$$\cos\theta = \frac{z}{r} = \frac{z}{\sqrt{x^2+y^2+z^2}}$$

$$\sin\phi = \frac{y}{\rho} = \frac{y}{\sqrt{x^2+y^2}}$$

$$\cos\phi = \frac{x}{\rho} = \frac{x}{\sqrt{x^2+y^2}}$$

Substituting all these gives:

$$B_x = \frac{10\sqrt{x^2+y^2}}{(x^2+y^2+z^2)} \cdot \frac{x}{\sqrt{x^2+y^2}} + \frac{\sqrt{x^2+y^2+z^2}}{(x^2+y^2+z^2)} \cdot \frac{z^2x}{\sqrt{x^2+y^2}} - \frac{\mathrm{y}}{\sqrt{x^2+y^2}}$$

$$B_x = \frac{10x}{x^2+y^2+z^2} + \frac{xz^2}{\sqrt{(x^2+y^2)(x^2+y^2+z^2)}} - \frac{\mathrm{y}}{\sqrt{x^2+y^2}}$$

$$B_y = \frac{10\sqrt{x^2+y^2}}{(x^2+y^2+z^2)} \cdot \frac{y}{\sqrt{x^2+y^2}} + \frac{\sqrt{x^2+y^2+z^2}}{(x^2+y^2+z^2)} \cdot \frac{z^2y}{\sqrt{x^2+y^2}} - \frac{\mathrm{y}}{\sqrt{x^2+y^2}}$$

$$B_y = \frac{10y}{x^2+y^2+z^2} + \frac{yz^2}{\sqrt{(x^2+y^2)(x^2+y^2+z^2)}} + \frac{\mathrm{y}}{\sqrt{x^2+y^2}}$$

$$B_z = \frac{10z}{(x^2+y^2+z^2)} - \frac{z\sqrt{x^2+y^2}}{\sqrt{x^2+y^2+z^2}}$$

$$\boldsymbol{B} = B_x\boldsymbol{a_x} + B_y\boldsymbol{a_y} + B_z\boldsymbol{a_z}$$

Where B_x, B_y, and B_z are as found before.

b. Finding $\boldsymbol{B}$ $(-3, 4, 0)$ and $(5,\ \frac{\pi}{2},\ -2)$:

At $(-3,4,0)$**,** $x = -3,\ \ y = 4$, and $z = 0$, so:

$$B_x = -\frac{30}{25} + 0 - \frac{4}{5} = -2$$

$$B_y = \frac{40}{25} + 0 - \frac{3}{5} = 1$$

$$B_z = 0 - 0 = 0$$

Thus:

$\boldsymbol{B} = -2\,\boldsymbol{a_x} + \boldsymbol{a_y}$

For spherical to cylindrical vector transformation:

$$\begin{bmatrix} B_\rho \\ B_\phi \\ B_z \end{bmatrix} = \begin{bmatrix} \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \\ \cos\theta & -\sin\theta & 0 \end{bmatrix} \begin{bmatrix} \frac{10}{\mathrm{r}} \\ \mathrm{r}\ \cos\theta \\ 1 \end{bmatrix}$$

or

$$B_\rho = \frac{10}{r}\sin\theta + rcos^2\theta$$

$$B_\phi = 1$$

$$B_z = \frac{10}{r}\cos\theta - r\sin\theta\cos\theta$$

But

$$r = \sqrt{\rho^2 + z^2}$$

And

$$\theta = tan^{-1}\frac{\rho}{z}.$$

Thus

$$\sin\theta = \frac{\rho}{\sqrt{\rho^2 + z^2}}$$

$$\cos\theta = \frac{z}{\sqrt{\rho^2 + z^2}}$$

$$B_\rho = \frac{10\rho}{\rho^2 + z^2} + \sqrt{\rho^2 + z^2}\,.\frac{z^2}{\rho^2 + z^2}$$

$$B_z = \frac{10z}{\rho^2 + z^2} - \sqrt{\rho^2 + z^2}\,.\frac{\rho z}{\rho^2 + z^2}$$

Hence

$$\boldsymbol{B} = \left(\frac{10\rho}{\rho^2 + z^2} + \frac{z^2}{\sqrt{\rho^2 + z^2}}\right)\boldsymbol{a_\rho} + \boldsymbol{a_\phi} + \left(\frac{10z}{\rho^2 + z^2} - \frac{\rho z}{\sqrt{\rho^2 + z^2}}\right)\boldsymbol{a_z}$$

At $(5, \frac{\pi}{2}, -2)$, $\rho = 5$, $\phi = \frac{\pi}{2}$, and $z = -2$

So

$$\boldsymbol{B} = \left(\frac{50}{29} + \frac{4}{\sqrt{29}}\right)\boldsymbol{a_\rho} + \boldsymbol{a}_\phi + \left(\frac{-20}{29} + \frac{10}{\sqrt{29}}\right)\boldsymbol{a_z}$$

$$= 2.467\boldsymbol{a_\rho} + \boldsymbol{a}_\phi + 1.167\boldsymbol{a_z}$$

Note that at $(-3, 4, 0)$

$$|\boldsymbol{B}(x, y, z)| = |\boldsymbol{B}(\rho, \phi, z)| = |\boldsymbol{B}(\rho, \theta, \phi)| = 2.907$$

2. a.

$$\boldsymbol{E} \times \boldsymbol{F} = \begin{vmatrix} \boldsymbol{a}_\rho & \boldsymbol{a}_\phi & \boldsymbol{a_z} \\ -5 & 10 & 3 \\ 1 & 2 & -6 \end{vmatrix}$$

$$\boldsymbol{E} \times \boldsymbol{F} = (-60 - 6)\,\boldsymbol{a_\rho} + (3 - 30)\,\boldsymbol{a_\phi} + (-10 - 10)\,\boldsymbol{a_z} = (-66, -27, -20)$$

$$|\boldsymbol{E} \times \boldsymbol{F}| = \sqrt{66^2 + 27^2 + 20^2} = 74.06$$

b. The vector component of $\boldsymbol{E}$ at $P(5, \frac{\pi}{2}, 3)$ parallel to the line $x = 2$, $z = 3$:

$(\boldsymbol{E}.\boldsymbol{a}_y)\,\boldsymbol{a}_y$

But at $P(5, \frac{\pi}{2}, 3)$

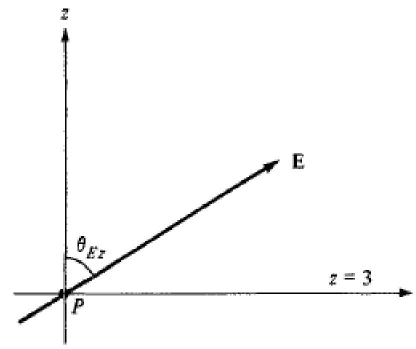
$$\boldsymbol{a}_y = \sin\phi\,\boldsymbol{a}_\rho + \cos\phi\,\boldsymbol{a}_\phi$$

$$\boldsymbol{a}_y = \sin\frac{\pi}{2}\boldsymbol{a}_\rho + \cos\frac{\pi}{2}\boldsymbol{a}_\phi = \boldsymbol{a}_\rho$$

Therefore

$$(\boldsymbol{E}.\boldsymbol{a_y})\,\boldsymbol{a}_{\boldsymbol{y}} = (\boldsymbol{E}.\boldsymbol{a_\rho})\,\boldsymbol{a_\rho} = -5\boldsymbol{a_\rho}\ \boldsymbol{(}or\ -5\boldsymbol{a_y}\boldsymbol{)}$$

c. The angle $\boldsymbol{E}$ makes with the surface $z = 3$ at P. Utilizing the fact that the z-axis is normal to the surface $z = 3$, the angle between the z-axis and $\boldsymbol{E}$, as shown in the following figure, can be found using the dot product:



$\boldsymbol{E}.\boldsymbol{a}_z = |\mathbf{E}|(1)\cos\theta_{Ez} \rightarrow 3 = \sqrt{134}\cos\theta_{Ez}$

$\cos\theta_{Ez} = \frac{3}{\sqrt{134}} = 0.2592 \rightarrow \theta_{Ez} = 74.98°$

Hence, the angle between $z = 3$ and $\boldsymbol{E}$ is:

$90° - \theta_{Ez} = 15.02°$

3. a. A unit vector along $\boldsymbol{H}$**:**

At P**,** $\rho = 2, \phi = 30°, \text{z} = -1$

$\boldsymbol{H} = -5\,\boldsymbol{a_\rho} + 2\cos 30°\,\boldsymbol{a_\phi} - 6\boldsymbol{a_z}$

$\boldsymbol{H} = -5\,\boldsymbol{a_\rho} + 1.732\boldsymbol{a_\phi} - 6\,\boldsymbol{a_z}$

$\boldsymbol{a_n} = \frac{(-5,1.732,-6)}{\sqrt{5^2 + 1.732^2 + 6^2}} = -0.625\,\boldsymbol{a_\rho} + 0.2165\,\boldsymbol{a_\phi} - 0.75\,\boldsymbol{a_z}$

The component of $\boldsymbol{H}$ parallel to $\boldsymbol{a_x}$**:**

$H_x = H_\rho\cos\phi - \ H_\phi\cos\phi = -5\rho\ \sin\phi\cos\phi - \rho z\cos\phi\sin\phi$

Or

P at $\rho = 2, \phi = 30°, \text{z} = -1$:]

$\boldsymbol{H} = H_x\,\boldsymbol{a_x} = (-10\sin 30°\cos 30° + 2\sin 30°\cos 30°\ \,)\,\boldsymbol{a_x}$

$\boldsymbol{H} = -3.4641\boldsymbol{a_x}$

The component of $\boldsymbol{H}$ normal to $\rho = 2$:

$\boldsymbol{H_n} = H_\rho\boldsymbol{a}_\rho = -0.625\,\boldsymbol{a_\rho}$

The component of $\boldsymbol{H}$ tangential to $\phi = 30°$:

$\boldsymbol{H_t} = H_\rho\,\boldsymbol{a}_\rho - H_z\,\boldsymbol{a}_z = -0.625\,\boldsymbol{a_\rho} - 0.75\,\boldsymbol{a_z}$